Given that the input is $ae^{s_1 t}$ to a LTI system with impulse response, H(s) what is the output $y(t)$? $y(t) = \alpha H(s_1)e^{s_1 t}$; t_r = rise time (10% to 90% of final value); t_p = peak time (time to peak); M_p = overshoot (diff of peak to steady state, % of final value); ζ = damp ratio (big = more damp); ω_n = undamped nat freq σ = damp factor (decay rate); ω_d = damped nat freq; t_r = 1.8/ω_n; t_p = π/ ω_d; t_s = 4.6/$\zeta\omega_n$ = 4.6/σ (for 1% settling time)

$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$; $\zeta = -\frac{\ln(\ \ p)}{\sqrt{\pi^2+\ln^2 M_p}}$; $\theta = \mathrm{Sin}^{-1}\zeta$; $\omega_d = \omega_n\sqrt{1-\zeta^2}$; $H(s) = \frac{K\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$ (std form 2order TF); $\sigma = \zeta\omega_n$; Pole closer to origin=The system slower.;

Zero closer to origin=system faster; Pole / Zero in LHP far from origin=insignificant; Faster Response: Smaller r_t, Shorter t_s; Slower Response: Larger r_t, Longer t_s; Odd # zeros in RHP = undershoot; Even # zeros in RHP = no undershoot; Unstable = {any poles RHP; > thn one point at origin; pole im axis but not origin (marg stable)} Unity feedbach = nothing in feedback loop; Routh array → can know if system is stable, and how many poles there are in RHP. If ANY neg coeffs = unstable.

If all pos, could still be RHP. #poles in RHP = #sign changes in first row of routh array. Ex: 1,3,-7,8 = 2 sign changes, from + to – then - to +. To construct array, use leftmost column, and two rows above target. Use column directly to the right of target value. $\frac{(\mathrm{bottom_{Left}})(\mathrm{Top_{Right}})-(\mathrm{To\ \ Left})(\mathrm{Bottom_{Right}})}{\mathrm{Bottom_{Left}}}$; Final value theorem: $y(\infty) = \lim_{s\to 0} sH(s)$

DC Gain Theorem (assuming input is unit step, if apply unit step, what will final value be?): DC Gain = $\lim_{s\to 0} s(H(s) * \frac{1}{s})$; Initial Value Theorem: $\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s)$

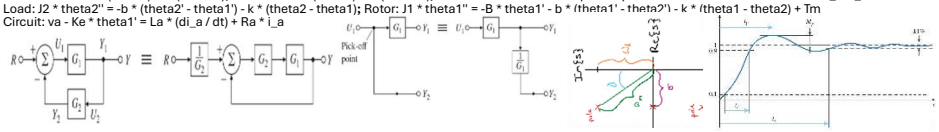
$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt = f(t_0)$ (integrating an impulse multiplied by a function = sifting property of an impulse) ; Step response: H(s)=$\frac{Y(s)}{U(s)} = \frac{Y(s)}{1/s} \rightarrow y(t) =$ step response

Ex imp resp: $H(s) = \frac{10s}{(s+4)(s+3)} = \frac{40}{s+4} + \frac{-30}{s+3} \rightarrow \mathrm{inL} \rightarrow -30e^{-3t} + 40e^{-4t} = h(t)$(impulse response!); DC motor: output speed given voltage: $\frac{\Omega(s)}{V(s)} = \frac{K}{Ts+1}$

non-minimum phase system = one or more zeros in the right-half plane (RHP); v = ẋ; a = ẋ; d/dt i_a → usually goes to 0
Law of motors: F = BliN. F → mech force. B → mag flux density. l → length of conductor exposed to mag field. i → elec current through conductor
Law of generators: e = Blv V. e → induced voltage. B → mag flux density. l → length of conductor exposed to mag field. v → velocity of conductor.

Sum of forces = ma; $T_{\mathrm{motorFriction}} = B\dot{\theta}$; $T_{\mathrm{motor}} = k_t i_a$; k_t = motor torque constant; $V_{\mathrm{emf}} = k_e\dot{\theta}$; k_e = back emf constant; $F_{\mathrm{spring}} = -kx$; $F_{\mathrm{damper}} = b\dot{x}$

newtons law for rotational motion: $T = J\ddot{\theta}$; J=moment of inertia; $\ddot{\theta}$ = rotational accel; Omp amp assumptions: $V_p = V_n$; $I_p = I_n = 0$; Pos = leader; neg = follower
Capacitor i = C dv/dt; Inductor: V = L di/dt; row of zeros in routh array = roots symmetrically about the origin, imaginary or real; Settling time = 5 time constants (T)
Adding a pole at s = 0 changes a system from one that reacts to current inputs into one that accumulates inputs over time. Adds integrator into controller.
If nodes have same start point and go into same summer, they can be combined into a single sum. ; Inv lap of transfer fun = impulse response
Steady state error: $y_{ss} = \lim sY(s)$, remember to include another 1/s if it is steady state error to a step response! ; U(t)→invlap→1/s ; Up on IM = faster oscillations
Gain of a single loop **NEG** feedback system is given by the forward gain divided by the sum of 1 plus the loop gain.; R on real=faster growth. Left on real=decay faster
{Re{z} < 4σ (LHP): M_p up, t_r down, t_s no effect; Re{p}<4σ(LHP): M_P down, t_r up, t_s up}; poly long div on HP prime →toolbox, poly, algebra, quotient/remainder;
m1x1'' = -k1x1 - k2(x1 - x2) - b1(x1' - x2'); m2x2'' = F - k2(x2 - x1) - b1(x2' - x1'); Relation between the output torque and the armature current: Tm = K_t * i_a
Load: J2 * theta2'' = -b * (theta2' - theta1') - k * (theta2 - theta1); Rotor: J1 * theta1'' = -B * theta1' - b * (theta1' - theta2') - k * (theta1 - theta2) + Tm
Circuit: va - Ke * theta1' = La * (di_a / dt) + Ra * i_a



(V_in - V_-) / R_in = (V_- - V_out) / R_f ;; V_- = (R_f / (R_in + R_f)) * V_in + (R_in / (R_in + R_f)) * V_out ;; V_out = (10^7 / (s + 1)) * [V_+ - V_-]
= (10^7 / (s + 1)) * (V_+ - (R_f / (R_in + R_f)) * V_in - (R_in / (R_in + R_f)) * V_out) ;; = -(10^7 / (s + 1)) * ((R_f / (R_in + R_f)) * V_in + (R_in / (R_in + R_f)) * V_out)
V_out / V_in = (-10^7 * (R_f / (R_in + R_f))) / (s + 1 + 10^7 * (R_in / (R_in + R_f)))
2nd: (V_in - 0) / R_in = (0 - V) / R_2 = (V - V_out) / R_1 + C * d/dt(V - V_out) ;; V = -(R_2 / R_in) * V_in ;;
V_in / R_in = (-(R_2 / R_in) * V_in - V_out) / R_1 + C * d/dt(-(R_2 / R_in) * V_in - V_out)
= (1 / R_1) * (-(R_2 / R_in) * V_in - V_out) + C * (-(R_2 / R_in) * V_in' - V_out') ;; (1 / R_in) * (1 + (R_2 / R_1)) * V_in + (1 / R_in) * R_2 * C * V_in' = -(1 / R_1) * V_out - C * V_out'
V_out / V_in = -(R_1 / R_in) * ((R_2 * C * s + 1 + (R_2 / R_1)) / (R_1 * C * s + 1)) = -(R_2 / R_in) * ((s + (1 / (R_2 * C)) + (1 / (R_1 * C))) / (s + (1 / (R_1 * C))))

